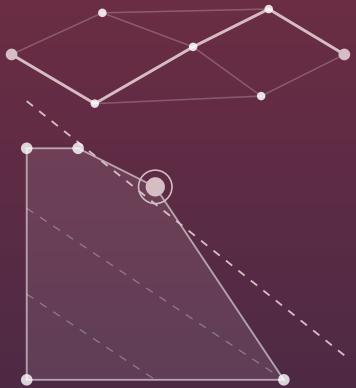
Operations Research

Lecture 5 · Semester 5



المدرسة الوطنية العليا لتكنولوجيا الأنظمة المستقلة

NHSAST

NATIONAL HIGHER SCHOOL OF AUTONOMOUS SYSTEMS TECHNOLOGY

Duality

The price of the resources

NATIONAL HIGHER SCHOOL OF MATHEMATICS
المدرسة الوطنية العليا في الرياضيات

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Today's journey: six parts

1. What are 25 and 5?
2. The dual problem
3. Weak and strong duality
4. Shadow prices, decisions
5. Complementary slackness
6. The dual of any problem

1. What are the numbers 25 and 5?
2. The dual problem
3. Weak and strong duality
4. Shadow prices and decisions
5. Complementary slackness
6. The dual of any problem

Goal understand and use the value of each resource, read in the last tableau.

Part 1 of 6 · minutes 2–17

What are the numbers 25 and 5?

1 What are 25 and 5?
2 The dual problem
3 Weak and strong duality
4 Shadow prices, decisions
5 Complementary slackness
6 The dual of any problem

In this part, you learn to

- find the numbers 25 and 5 in the last tableau, and use them: multiply and add;
- see why this proves that the plan is optimal;
- understand what they are: the value of one more hour.

Where we are: two numbers in the last z-line

Drone workshop (profit in thousand DA per week):

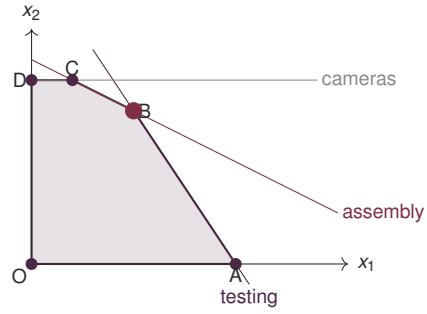
$$\max z = 40x_1 + 60x_2$$

$$\text{s.t. } x_1 + 2x_2 \le 40 \quad \text{(assembly hours)}$$

$$3x_1 + 2x_2 \le 60 \quad \text{(testing hours)}$$

$$x_2 \le 18 \quad \text{(cameras)}, \qquad x \ge 0$$

Best plan: $\mathrm{B} = (10, 15)$, $z^* = 1300$.



Lecture 3, last z-line: $z = 1300 - 25\,s_1 - 5\,s_2$.

Lecture 2, the proof: $25(x_1 + 2x_2) + 5(3x_1 + 2x_2) = 40x_1 + 60x_2 \le 25 \times 40 + 5 \times 60 = 1300$.

Today how do we use 25 and 5? Why does it work? What are they?

How to use them: multiply and add

		x_1	x_2	s_1	s_2	s_3	RHS
last z-line	z	0	0	25	5	0	1300

slack	its constraint	multiply by	result
s_1	assembly: $x_1 + 2x_2 \leq 40$	25	$25x_1 + 50x_2 \leq 1000$
s_2	testing: $3x_1 + 2x_2 \leq 60$	5	$15x_1 + 10x_2 \leq 300$
s_3	cameras: $x_2 \leq 18$	0	not used
	add		$\mathbf{40x_1 + 60x_2 \leq 1300}$

Rule multiply constraint i by the number under its slack s_i in the last z-line, and add. Every plan earns ≤ 1300; B earns 1300: B is optimal.

Why it works: the last dictionary is the proof

Last dictionary (Lecture 3): $z = 1300 - 25\,s_1 - 5\,s_2$.

The slacks are the unused hours:

$s_1 = 40 - x_1 - 2x_2, \qquad s_2 = 60 - 3x_1 - 2x_2.$

Replace them:

$$z = 1300 - 25\,(40 - x_1 - 2x_2) - 5\,(60 - 3x_1 - 2x_2)$$

$$= \underbrace{1300 - 1000 - 300}_{0} + 25\,(x_1 + 2x_2) + 5\,(3x_1 + 2x_2).$$

So $40x_1 + 60x_2 = 25\,(x_1 + 2x_2) + 5\,(3x_1 + 2x_2)$: the line of Lecture 2.

Why ≤ 1300 $s_1 \geq 0$ and $s_2 \geq 0$ for every feasible plan, so

$$z = 1300 - \underbrace{25\,s_1}_{\geq 0} - \underbrace{5\,s_2}_{\geq 0} \leq 1300.$$

The long line of Lecture 2 is the last dictionary, written with the constraints instead of the slacks.

What are 25 and 5? The value of the hours

Give the value 25 to an assembly hour and 5 to a testing hour.

- Drone A (1 assembly h, 3 testing h): $25 + 3 \times 5 = 40 =$ its profit.
- Drone B (2 assembly h, 2 testing h): $2 \times 25 + 2 \times 5 = 60 =$ its profit.

So, for every plan:
profit $= 25\times$ (assembly h used) $+ 5\times$ (testing h used).

plan	assembly h	testing h	$25 \times$ as. $+ 5 \times$ te.
A $= (20, 0)$	20	60	$500 + 300 = 800$
$(10, 10)$	30	50	$750 + 250 = 1000$
C $= (4, 18)$	40	48	$1000 + 240 = 1240$
B $= (10, 15)$	40	60	$1000 + 300 =$ **1300**

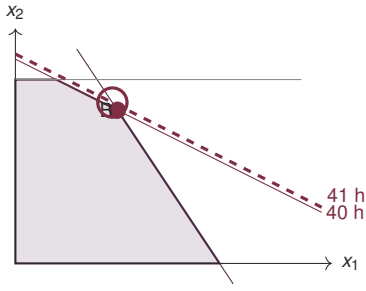
Each value is the real profit of the plan ✓

Meaning only 40 and 60 hours: profit $\leq 25 \times 40 + 5 \times 60 = 1300$, reached when all the hours are used. You cannot earn more than what your resources are worth.

What do they mean? One more hour

change	new best plan	new z^*	change of z^*
41 assembly hours	$(9.5,\ 15.75)$	1325	$+25$
39 assembly hours	$(10.5,\ 14.25)$	1275	-25
61 testing hours	$(10.5,\ 14.75)$	1305	$+5$
19 cameras	$(10,\ 15)$	1300	0

So 25 is the extra profit given by one more assembly hour; 5 by one more testing hour; 0 by one more camera (3 are unused).



Not a cost, not an income: what one hour is worth to the workshop. Its name: shadow price (Part 4).

The dual problem

1 What are 25 and 5?
2 The dual problem
3 Weak and strong duality
4 Shadow prices, decisions
5 Complementary slackness
6 The dual of any problem

In this part, you learn to

- look at the workshop from the side of someone who rents its resources;
- write this new problem with variables y_1, y_2, y_3: the dual problem;
- write the dual in canonical form: rows become columns (Definition 5.1).

Let us consider the following situation

An investor wants to rent all the resources of the workshop for one week: 40 assembly hours, 60 testing hours and 18 cameras.

- He offers a rent for each unit (thousand DA): y_1 per assembly hour, y_2 per testing hour, y_3 per camera, $y_1, y_2, y_3 \geq 0$.
- The owner accepts only if renting pays at least as much as making drones. The hours of one drone A must bring at least 40: $y_1 + 3y_2 \geq 40$. For drone B: $2y_1 + 2y_2 + y_3 \geq 60$.
- The investor pays $40y_1 + 60y_2 + 18y_3$ in total, and wants to pay as little as possible.

The investor's problem

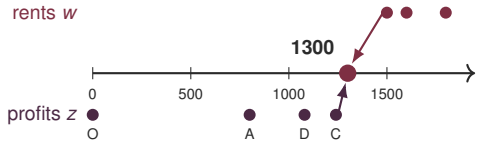
$$\begin{aligned} \min\ & w = 40y_1 + 60y_2 + 18y_3 \\ \text{s.t. } & y_1 + 3y_2 \geq 40 \quad \text{(drone A)} \\ & 2y_1 + 2y_2 + y_3 \geq 60 \quad \text{(drone B)} \\ & y_1, y_2, y_3 \geq 0 \end{aligned}$$

Variables: the rents y_1, y_2, y_3. One constraint per drone, one variable per resource.

The best rent: this is the dual problem

y_1	y_2	y_3	drone A	drone B	rent w
0	30	0	90 ✓	60 ✓	1800
40	0	0	40 ✓	80 ✓	1600
30	5	0	45 ✓	70 ✓	1500
20	0	0	20 ×	40 ×	refused
25	5	0	40 ✓	60 ✓	1300

The best rent: $y = (25, 5, 0)$, $w = 1300$: 25 and 5 again, and the same value as the best profit.



Profits go up to 1300; acceptable rents come down to 1300.

Names the investor's problem is the dual problem; the drone problem is the primal problem.

Definition 5.1: the primal and its dual

Definition 5.1 — Dual problem

Primal (m constraints, n variables) $\qquad$ **Dual** (n constraints, m variables)

$$\max\ z = c^\top x \ \text{ s.t. } \ Ax \le b,\ \ x \ge 0 \quad \longleftrightarrow \quad \min\ w = b^\top y \ \text{ s.t. } \ A^\top y \ge c,\ \ y \ge 0$$

Primal (the drone workshop):

$$\begin{aligned} &\max\ 40x_1 + 60x_2 \\ &\text{s.t. } x_1 + 2x_2 \le 40,\ \ 3x_1 + 2x_2 \le 60 \\ &\qquad x_2 \le 18,\ \ x \ge 0 \end{aligned}$$

Dual (the rents):

$$\begin{aligned} &\min\ 40y_1 + 60y_2 + 18y_3 \\ &\text{s.t. } y_1 + 3y_2 \ge 40 \\ &\qquad 2y_1 + 2y_2 + y_3 \ge 60,\ \ y \ge 0 \end{aligned}$$

- One dual variable y_i per primal constraint; one dual constraint per primal variable.
- b becomes the objective, c the right-hand side, A is transposed; max $\to$ min, "$\le$" $\to$ "$\ge$".

Example 5.1: rows become columns

	x_1	x_2	RHS	rent
assembly	1	2	40	y_1
testing	3	2	60	y_2
cameras	0	1	18	y_3
profit	40	60		

The primal reads the table by rows. The dual reads it by columns:

- Column x_1: $1y_1 + 3y_2 + 0y_3 \geq 40$.
- Column x_2: $2y_1 + 2y_2 + 1y_3 \geq 60$.
- Column RHS: $\min\ w = 40y_1 + 60y_2 + 18y_3$.
- And $y_1, y_2, y_3 \geq 0$.

Check 2 variables, 3 constraints $\rightarrow$ 3 variables, 2 constraints.

Why "$\geq$" in the dual constraints?

Multiply the constraints by $y_1, y_2, y_3 \geq 0$ and add (as in Part 1):

$$(y_1 + 3y_2)\, x_1 + (2y_1 + 2y_2 + y_3)\, x_2 \;\leq\; 40y_1 + 60y_2 + 18y_3.$$

$y = (40, 0, 0)$**:** $40x_1 + 80x_2 \leq 1600$.
In front of x_1: $40 \geq 40$ ✓; of x_2: $80 \geq 60$ ✓
Since $x_2 \geq 0$: $60x_2 \leq 80x_2$, so
profit $= 40x_1 + 60x_2 \leq 40x_1 + 80x_2 \leq 1600$ ✓

$y = (20, 0, 0)$**:** $20x_1 + 40x_2 \leq 800$.
In front of x_1: $20 < 40$ ×; of x_2: $40 < 60$ ×
The left side is smaller than the profit: it says nothing about the profit. Indeed B earns $1300 > 800$ ×

Rule the dual constraints ask: in front of each x_j, at least its profit c_j. Then profit $\leq$ left side $\leq$ rent. So every acceptable rent is a bound, and the dual looks for the best bound.

Your turn: the dual of the control boards

TD 1 (profit in thousand DA per day):

$$\begin{aligned} \max\ & z = 12x_1 + 20x_2 \\ \text{s.t. } & 2x_1 + 4x_2 \le 480 \quad \text{(machine)} \\ & 3x_1 + 2x_2 \le 360 \quad \text{(test bench)} \\ & x_2 \le 100 \quad \text{(sensors)}, \quad x \ge 0 \end{aligned}$$

Your turn · 2 min write the dual.

Answer. y_1, y_2, y_3: values of a machine minute, a test bench minute, a sensor:

$$\begin{aligned} \min\ & w = 480y_1 + 360y_2 + 100y_3 \\ \text{s.t. } & 2y_1 + 3y_2 \ge 12 \quad \text{(Basic board)} \\ & 4y_1 + 2y_2 + y_3 \ge 20 \quad \text{(Pro board)} \\ & y_1, y_2, y_3 \ge 0 \end{aligned}$$

TD 2: last z-line $z = 2520 - \frac{9}{2}s_1 - s_2$, so $y = (\frac{9}{2}, 1, 0)$ and $w = 2160 + 360 = 2520$ ✓

The dual of the dual is the primal

1. Write the dual in canonical form (a max, with "$\leq$"):

$$\min\ b^\top y,\ A^\top y \geq c \quad \Longleftrightarrow \quad \max\,(-b)^\top y,\ \ (-A^\top)\,y \leq -c,\ \ y \geq 0.$$

2. Apply Definition 5.1 (variables x, matrix transposed again):

$$\min\,(-c)^\top x,\ \ (-A)\,x \geq -b,\ \ x \geq 0.$$

3. Multiply by -1: $\max\ c^\top x,\ Ax \leq b,\ x \geq 0$: the primal.

So primal and dual are a pair: each one is the dual of the other. We may start from either of them.

Part 3 of 6 · minutes 37–50

Weak and strong duality

1 What are 25 and 5?
2 The dual problem
3 **Weak and strong duality**
4 Shadow prices, decisions
5 Complementary slackness
6 The dual of any problem

In this part, you learn to

- prove that every plan is below every bound (weak duality, Key fact 5.1);
- use a dual solution as a certificate of optimality;
- know that the best values are equal (strong duality, Key fact 5.2) and all the cases.

Key fact 5.1: weak duality

Key fact 5.1 — Weak duality

If x is feasible for the primal and y is feasible for the dual, then $c^\top x \le b^\top y$ $(z \le w)$.

Why? Use $x \ge 0$ with $A^\top y \ge c$, then $y \ge 0$ with $Ax \le b$:

$$z = c^\top x \underset{x\ge 0}{\le} (A^\top y)^\top x = y^\top (Ax) \underset{y\ge 0}{\le} y^\top b = w.$$

Example. $x = (10, 10)$: $z = 1000$. $y = (30, 5, 0)$: $w = 1500$. Indeed $1000 \le 1500$.

In words the value of any plan is at most the rent of any acceptable prices: the bound of Part 2.

Consequences of weak duality

1. **A bound for free.** Any dual feasible y gives an upper bound on the best profit.
2. **A certificate.** If x and y are feasible and $z = w$, then x and y are both optimal. Example: $x = (10, 15)$ and $y = (25, 5, 0)$ both give 1300: no plan can beat 1300, no prices can go below 1300.
3. **Unbounded primal $\Rightarrow$ infeasible dual.** If $z \to +\infty$, no y can give a bound.
4. **Unbounded dual $\Rightarrow$ infeasible primal.** If $w \to -\infty$, no x can be below it.

Use to prove that a plan is optimal, show one dual feasible y with the same value.

Key fact 5.2: strong duality

Key fact 5.2 — Strong duality

If the primal has an optimal solution, the dual has one too, and the optimal values are equal: $z^* = w^*$.

Why? The last tableau gives y. Its z-line has no negative number:

- under s_1, s_2, s_3: $25, 5, 0 \geq 0$: this is $y \geq 0$;
- under x_1, x_2: $0, 0 \geq 0$: this is $A^\top y - c \geq 0$: $25 + 15 - 40 = 0$, $50 + 10 + 0 - 60 = 0$;
- the RHS: $1300 = b^\top y$.

So y is dual feasible with $w = z^*$: by the certificate, it is optimal.

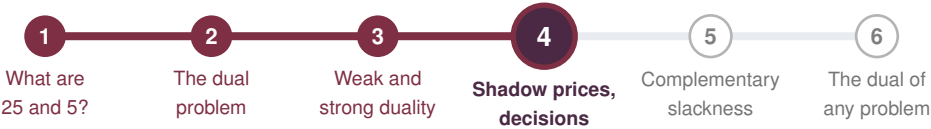
T3	x_1	x_2	s_1	s_2	s_3	RHS
x_1	1	0	$-\frac{1}{2}$	$\frac{1}{2}$	0	10
s_3	0	0	$-\frac{3}{4}$	$\frac{1}{4}$	1	3
x_2	0	1	$\frac{3}{4}$	$-\frac{1}{4}$	0	15
z	0	0	25	5	0	1300
			y_1	y_2	y_3	

All the cases for a primal–dual pair

Primal	Dual: optimal	Dual: unbounded	Dual: infeasible
optimal	✓ $z^* = w^*$	impossible	impossible
unbounded	impossible	impossible	✓
infeasible	impossible	✓	✓ possible

- Optimal: equal values (Key fact 5.2).
- Unbounded on one side $\Rightarrow$ infeasible on the other (weak duality).
- The other boxes contradict weak or strong duality.
- Both infeasible can happen:
  $\max\ 2x_1 - x_2,\ x_1 - x_2 \leq 1,$
  $-x_1 + x_2 \leq -2$
  ($x_1 - x_2$ cannot be ≤ 1 and ≥ 2; its dual has the same trouble).

Shadow prices and decisions



In this part, you learn to

- read the dual solution in any last tableau, with slacks and surpluses (Key fact 5.3);
- check the shadow prices by solving again (Definition 5.2);
- use them to decide: buy or sell a resource? make a new product?

Key fact 5.3: the dual solution in any last tableau

Key fact 5.3 — Reading the dual solution

In the last z-line (a min problem is solved as $\max\ -z$), let $d_i \geq 0$ be the number under the slack or surplus of constraint i. Then $y_i = d_i$ if constraint i is natural ($\leq$ in a max, $\geq$ in a min), and $y_i = -d_i$ if it is strange ($\geq$ in a max, $\leq$ in a min).

problem	last z-line	dual solution
drone workshop (max, all $\leq$)	25, 5, 0 under s_1, s_2, s_3	$y = (25, 5, 0)$
spare-part boxes (min, all $\geq$; Lecture 4)	$\frac{20}{3}, \frac{50}{3}$ under e_1, e_2	$y = (\frac{20}{3}, \frac{50}{3})$
drones + order $x_2 \geq 16$ (max; Lecture 4, Q4)	40 under s_1, 20 under e_4	$y_1 = 40,\ y_4 = -20$

For the proof turn every constraint in the direction of the bound ($\leq$ for a max, $\geq$ for a min), multiply by d_i and add. An "$=$" constraint has no slack: use Method 5.1.

Shadow prices: checks by solving again

Definition 5.2 — Shadow price

The shadow price of constraint i is the change of the best value when its right-hand side b_i increases by one unit (for small changes). It is equal to y_i.

problem	change	new best plan	new best value	change
drone workshop	41 assembly hours	$(9.5,\ 15.75)$	profit 1325	$+25 = y_1$ ✓
drone workshop	19 cameras	$(10,\ 15)$	profit 1300	$0 = y_3$ ✓
spare-part boxes	9 motors needed	$(\frac{11}{3},\ \frac{5}{3})$	cost 176.67	$+\frac{20}{3} = y_1$ ✓
drones + order	order of 17 drones B	$(6,\ 17)$	profit 1260	$-20 = y_4$ ✓

Signs a natural constraint is a resource or a need: $y_i \geq 0$. A strange constraint is a limit that hurts: an extra drone B ordered costs 20 of profit.

How far are these values valid? Sensitivity analysis, Lecture 6.

Using the prices: should we buy more?

A partner offers extra resources. Compare the offer with the shadow price:

- One more assembly hour for 20 thousand DA?
  It brings 25: yes, gain $25 - 20 = 5$.
- One more testing hour for 8?
  It brings 5: no, loss $5 - 8 = -3$.
- One more camera for 1?
  It brings 0: no.
- **Selling** one testing hour to the partner: accept only a price above 5.

Rule of thumb a resource is worth its shadow price: buy below it, sell above it. A resource that is not fully used is worth 0.

Pricing a new product: drone C

A new drone C needs 1 assembly hour, 1 testing hour and 1 camera. Its profit is p.

- Value of its resources at the shadow prices: $1 \times 25 + 1 \times 5 + 1 \times 0 = 30$.
- If $p > 30$, making C is worth it: $p = 35$ gives the new best plan $(10, 12, 6)$, $z^* = 1330$.
- If $p < 30$, do not make C: $p = 28$ gives the same plan B, $z^* = 1300$.

Pricing a product is worth making only if its profit is more than the value of its resources at the shadow prices. This is the dual constraint of the new product, not satisfied.

Part 5 of 6 · minutes 63–75

Complementary slackness

1 What are 25 and 5? — 2 The dual problem — 3 Weak and strong duality — 4 Shadow prices, decisions — **5 Complementary slackness** — 6 The dual of any problem

In this part, you learn to

- state complementary slackness (Key fact 5.4);
- find the dual solution from a primal solution, or show that a plan is not optimal (Method 5.1);
- see it on the picture.

Key fact 5.4: complementary slackness

Key fact 5.4 — Complementary slackness

A primal feasible x and a dual feasible y are both optimal if and only if

1. for each constraint i: $y_i = 0$ or constraint i is binding ($s_i = 0$);
2. for each variable j: $x_j = 0$ or dual constraint j is an equality.

- **Why?** In the chain $c^\top x \le y^\top Ax \le y^\top b$, optimal means $z = w$: both "$\le$" are "$=$". The second one is $\sum_i y_i s_i = 0$, the first one is $\sum_j x_j(\text{dual surplus})_j = 0$: sums of products ≥ 0, so each product is 0.
- **In words:** a resource that is not fully used has price 0; a product that is made earns exactly the value of its resources.
- Drone workshop: $s_3 = 3 > 0$ so $y_3 = 0$ ✓; $x_1, x_2 > 0$ so both dual constraints are equalities ✓

Method 5.1: from the plan to the prices

Method 5.1 — Test a plan x

1. Compute the slacks: if $s_i > 0$, set $y_i = 0$.
2. For each $x_j > 0$: write dual constraint j as an equation.
3. Solve for y.
4. If y is dual feasible: x and y are optimal. If the y found (unique) is not feasible: x is not optimal.

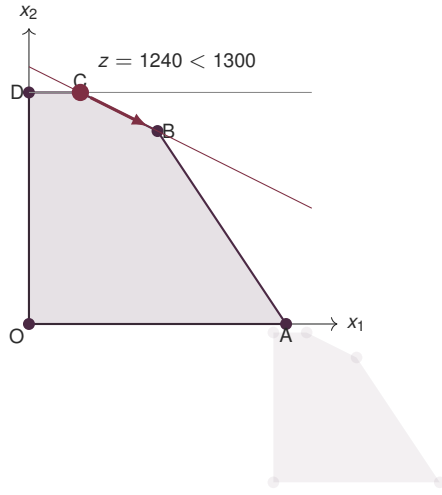
Test B $= (10, 15)$**.**

1. Slacks: $s_1 = 0$, $s_2 = 0$, $s_3 = 18 - 15 = 3 > 0$: $y_3 = 0$.
2. $x_1 > 0$: $y_1 + 3y_2 = 40$; $x_2 > 0$: $2y_1 + 2y_2 + y_3 = 60$.
3. $y_1 + 3y_2 = 40$, $y_1 + y_2 = 30$: $2y_2 = 10$, $y_2 = 5$, $y_1 = 25$.
4. $y = (25, 5, 0) \geq 0$, both dual constraints hold: B is optimal, $w = 1300 = z$ ✓

Is the corner C $= (4, 18)$ optimal?

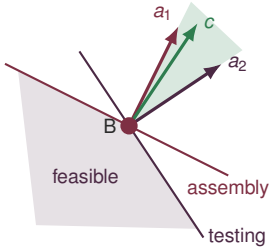
1. Slacks at C: assembly $4 + 36 = 40$: $s_1 = 0$; testing $12 + 36 = 48$: $s_2 = 12 > 0$, so $y_2 = 0$; cameras: $s_3 = 0$.
2. $x_1 > 0$: $y_1 + 3y_2 = 40$, so $y_1 = 40$.
3. $x_2 > 0$: $2y_1 + 2y_2 + y_3 = 60$, so $80 + y_3 = 60$: $y_3 = -20 < 0$ ×
4. The only possible prices are not feasible: C is not optimal.

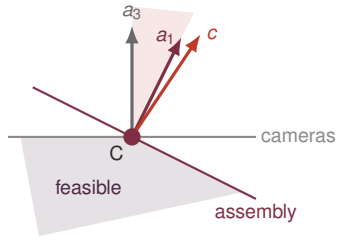
Meaning at C, a camera has a negative value: using fewer cameras (going from C towards B) increases the profit.



On the picture: the arrow c between the normals

Rows of A: $a_1 = (1, 2)$, $a_2 = (3, 2)$, $a_3 = (0, 1)$, perpendicular to the lines, pointing out.





At B: $c = (40, 60) = 25\, a_1 + 5\, a_2$: prices ≥ 0,
c is between a_1 and a_2: B optimal ✓

At C: $c = 40\, a_1 - 20\, a_3$: a negative price,
c is outside: C not optimal ×

Picture a corner is optimal when c is a combination of the normals of its binding lines with prices ≥ 0.

Part 6 of 6 · minutes 75–87

The dual of any problem

1. What are 25 and 5?
2. The dual problem
3. Weak and strong duality
4. Shadow prices, decisions
5. Complementary slackness
6. **The dual of any problem**

In this part, you learn to

- write the dual of a minimisation, of "$\geq$" and "$=$" constraints, of free variables (Rule 5.1);
- solve a small dual by drawing and compare with the primal;
- read the meaning of the dual of a cost problem.

Rule 5.1: the dual of any problem

Rule 5.1 — Writing the dual

Max problem		Min problem
constraint i: $\leq$	$\leftrightarrow$	$y_i \geq 0$
constraint i: $\geq$	$\leftrightarrow$	$y_i \leq 0$
constraint i: $=$	$\leftrightarrow$	y_i free
$x_j \geq 0$	$\leftrightarrow$	constraint j: $\geq$
$x_j \leq 0$	$\leftrightarrow$	constraint j: $\leq$
x_j free	$\leftrightarrow$	constraint j: $=$

- Always: rows $\leftrightarrow$ columns, $b \leftrightarrow c$, $A \rightarrow A^\top$.
- "Natural" $\leftrightarrow$ "natural" (Definition 5.1).
- "Strange" $\leftrightarrow$ "strange"; equality $\leftrightarrow$ free.
- A min primal: read the table from right to left.

Example 5.2: the dual of the spare-part boxes

Primal (Lectures 2 and 4, cost in thousand DA):

$$\begin{aligned} \min\ & z = 30x_1 + 40x_2 \\ \text{s.t. } & 2x_1 + x_2 \geq 8 \quad \text{(motors)} \\ & x_1 + 2x_2 \geq 7 \quad \text{(sensors)}, \quad x \geq 0 \end{aligned}$$

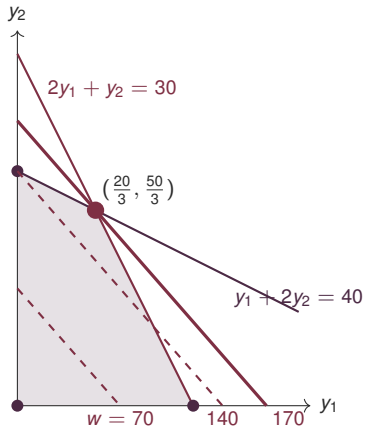
A min problem with natural "$\geq$" constraints: read Rule 5.1 from right to left.

Dual: y_1, y_2 = prices of a motor, a sensor:

$$\begin{aligned} \max\ & w = 8y_1 + 7y_2 \\ \text{s.t. } & 2y_1 + y_2 \leq 30 \quad \text{(box A)} \\ & y_1 + 2y_2 \leq 40 \quad \text{(box B)}, \quad y \geq 0 \end{aligned}$$

Meaning: a supplier sells motors and sensors separately. Its prices must not make a box more expensive than buying the parts; it wants the largest income for the 8 motors and 7 sensors we need.

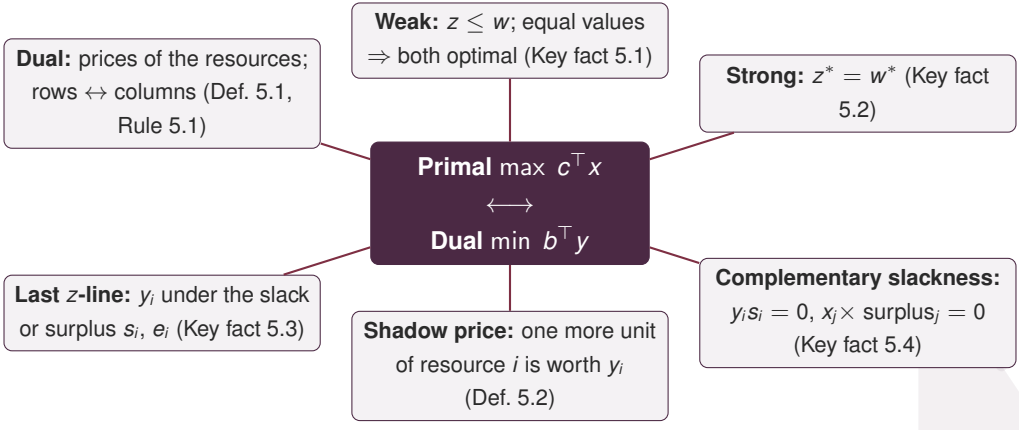
Solving the dual by drawing



- Corners: $(0,0)$: 0; $(15,0)$: 120; $(0,20)$: 140; $(\frac{20}{3}, \frac{50}{3})$: $\frac{160}{3} + \frac{350}{3} = 170$.
- Optimal prices: motor $\frac{20}{3} \approx 6.67$, sensor $\frac{50}{3} \approx 16.67$; $w^* = 170$.
- Primal: 3 boxes A and 2 boxes B cost $90 + 80 = 170$: $z^* = w^*$ ✓
- Complementary slackness: $x_1, x_2 > 0$, so both dual constraints are equalities: $2y_1 + y_2 = 30$, $y_1 + 2y_2 = 40$.

Meaning one more motor needed costs about 6.67 thousand DA; one more sensor about 16.67.

Summary: duality in one picture



TD 4: duality by hand. Next: Lecture 6, sensitivity analysis: how far can b and c change?